

Tuğçe Nur KARADENİZ

📍 Hatay, Türkiye ✉️ tugce.karadeniz@outlook.com 🌐 Website 📄 GitHub
🔖 ORCID

Education

- M.Sc. Fırat University** Elazığ, Türkiye
Sep 2023 – Jan 2026
Faculty of Technology
Department of Energy Systems Engineering
- GPA: 3.78/4.00
 - Thesis: *Damage Detection on Wind Turbine Blades Using YOLO-Based Deep Learning Approaches*
- B.Sc. Giresun University** Giresun, Türkiye
Sep 2012 – Jun 2017
Faculty of Engineering
Department of Energy Systems Engineering
- GPA: 3.02/4.00

Experience

- Ulaşım İç ve Dış Ticaret A.Ş.** Bursa, Türkiye
Jan 2020 – Sep 2022
Product Structure Engineer (R&D Department)
- Managed product structures, part codes, and engineering documentation using SAP ERP.
 - Prepared and maintained BOMs and technical instructions within the R&D department.
 - Collaborated with engineering and production teams on product development processes.
- Starwood Orman Ürünleri A.Ş.** Bursa, Türkiye
Jun 2018 – Dec 2018
Machine Operator
- Operated industrial energy systems including steam, hot oil, and cooling systems.
 - Performed equipment monitoring, maintenance, and efficiency improvement activities.

Publications

- Deep Learning-Based Detection of Power Transmission Lines Using YOLOv4 and YOLOv8** 2025
Tuğçe Nur Karadeniz, Sami Ekici, Engin Avcı
Scientific Reports
DOI: [10.1038/s41598-025-32200-w](https://doi.org/10.1038/s41598-025-32200-w)
- A-BiYOLOv9: An Attention-Guided YOLOv9 Model for Infrared-Based Wind Turbine Inspection** 2025
Sami Ekici, Murat Uyar, Tuğçe Nur Karadeniz
Applied Sciences
DOI: [10.3390/app152111840](https://doi.org/10.3390/app152111840)

Projects

Data Science Bootcamp Projects

May 2026

- Completed practical projects in data analysis, machine learning, and predictive modeling.
- Developed Python-based workflows including data cleaning, visualization, and model training.
- Worked with data science libraries and machine learning algorithms.

Skills

Programming / Markup Languages: Python, LaTeX

Machine Learning / Data Science: Pandas, NumPy, Scikit-learn, Data Preprocessing, Feature Engineering, Data Visualization

Deep Learning / Computer Vision: YOLOv4, YOLOv8, YOLOv9, Object Detection, Image Processing, OpenCV

Tools: Jupyter Notebook, Google Colab, Git, GitHub, Overleaf

Engineering: Renewable Energy Systems, Wind Energy Systems, Energy Efficiency

Languages: Turkish (native), English (A2)

Certificates

Data Science Bootcamp — May 2026

ISO 14001:2015 Environmental Management System, ISO 50001:2018 Energy Management System, Sustainability Fundamentals — IMES OSB Excellence Center, Oct 2023